Amir Moeini

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Research: Reinforcement learning and post-training of LLM agents (on-policy distillation from LLM-judge feedback, multi-turn agentic tasks); in-context RL for test-time scaling; safe RL; reasoning evaluation.

Education

University of Virginia | Ph.D. in Computer Science | Sep 2024 – Present
Sequential Intelligence Lab, Advisor: Shangtong Zhang | Charlottesville, VA

University of Virginia | M.S. in Computer Science (en route to Ph.D.) | Sep 2024 – Jan 2026

Sharif University of Technology | B.Sc. in Computer Engineering | Sep 2018 – Mar 2024
GPA: 19.1/20 (4/4) | Leave of absence to co-found a startup: 2021–2022 | Tehran, Iran

Publications & Preprints

- **Inductive Feedback for Mixed-Policy Distillation**
  Moeini A, Zhu HJ, Havir D, Zhang S
  Preprint | arxiv.org/abs/2609.35390
- **Latent Q-Barrier Shielding for Safe In-Context Reinforcement Learning**
  Kwon M*, **Moeini A***, Zhang S, Feng L
  Neural Information Processing Systems (**NeurIPS**) **2026** | arxiv.org/abs/2605.25267
- **MathlibLemma: Folklore Lemma Generation and Benchmark for Formal Mathematics**
  Liu X, Xie Z, **Moeini A**, Chen C, Liu SD, Meng Y, Zhang A, Zhang S
  International Conference on Machine Learning (**ICML**) **2026** | arxiv.org/abs/2602.02561
- **Safe In-Context Reinforcement Learning**
  Moeini A*, Kwon M*, Bozkurt AK, Motai Y, Chandra R, Feng L, Zhang S
  International Conference on Machine Learning (**ICML**) **2026** | arxiv.org/abs/2509.25582
- **Reward Is Enough: LLMs Are In-Context Reinforcement Learners**
  Song K*, **Moeini A***, Wang P, Gong L, Chandra R, Qi Y, Zhang S
  International Conference on Learning Representations (**ICLR**) **2026** | arxiv.org/abs/2506.06303
- **A Survey of In-Context Reinforcement Learning**
  Moeini A, Wang J, Beck J, Blaser E, Whiteson S, Chandra R, Zhang S
  Preprint | arxiv.org/abs/2502.07978
- **Mathador-LM: A Dynamic Benchmark for Mathematical Reasoning on Large Language Models**
  Kurtic E*, **Moeini A***, Alistarh D
  Empirical Methods in Natural Language Processing (**EMNLP**) **2024** | arxiv.org/abs/2406.12572

* equal contribution

Experience

Machine Learning Research Intern, Alquist Robotics, NYC — Jan 2026 – May 2026
RL Training for LLM-based Embodied Control

- Proposed and developed on-policy distillation conditioned on judge feedback, improving adherence to behavioral guidelines without a verifiable reward signal.

PhD Student, University of Virginia — Sep 2024 – Present
Reinforcement Learning, under supervision of Shangtong Zhang

- Doing research on reinforcement fine-tuning of large language models to enhance learning from feedback in multi-turn interactions.

Research Intern, IST Austria Apr 2024 – Aug 2024

LLM Quantization, under supervision of Dan Alistarh

- Improved GPTQ 4-bit (INT4 and NF4) quantization by optimizing scale and zero-point search, and by smoothing activation outliers, while maintaining compatibility with the Marlin kernel in the INT4 case.

Undergraduate Researcher, DML, Sharif University Mar 2023 – Dec 2023

RL for Single Viewpoint Active Perception, under supervision of Hamid R. Rabiee

- Implemented an on-policy RL algorithm to train a camera control policy with a vision transformer backbone that learns to look around along the most efficient trajectory to maximize information gain.

Quant Researcher, Wave Technologies, Isfahan Sep 2021 – Sep 2022

Time Series Forecasting

- Designed the forecasting stack end to end: Transformer and N-BEATS models trained with a curriculum on multi-node infrastructure, transfer learning across series, and conformal prediction for risk-controlled decisions.

Undergraduate Researcher, Onyxlab, Sharif University Jun 2021 – Sep 2021

End-to-End Persian Speech Recognition, under supervision of Abolfazl Motahari

- Masked speech modeling to learn voice token representations and a Persian language model to improve decoding.

Teaching and Service

- **Teaching Assistant**, Reinforcement Learning, University of Virginia (Fall 2025, Fall 2026)
- **Head Teaching Assistant**, Deep Learning (graduate) and Modern Information Retrieval, Sharif University: led teams of 25 and 20 TAs (2023)
- **Teaching Assistant**, Machine Learning, Artificial Intelligence, Medical Image Processing (graduate), Data Structures and Algorithms, Linear Algebra, Sharif University (2021–2023)
- **Reviewer**: NeurIPS 2026, ICML 2026, ICLR 2026–2027, AISTATS 2025–2026, AAMAS 2025–2026, RLC 2026; external reviewer: HSCC/ICCPS 2026

Honors and Awards

- Thinking Machines Lab Research Grant (2026)
- University of Virginia Computer Science Scholar Fellowship (2024)
- University of Alberta Graduate Recruitment Scholarship (2024)
- Eligible for Direct Admission to the Master's Program in Computer Engineering at Sharif University, having ranked in the **top 10% of the class** (2022)
- Member of Iran's **National Elites** Foundation (2018)
- Ranked **33rd** (top 0.02%) in the National University Entry Exam among 100,000+ students (2018)

Technical Skills

- **Methods:** GRPO, PPO, DAPO, RLOO, REINFORCE++, DPO, reward modeling (LLM-judge rewards), on-policy distillation, in-context RL, constrained RL
- **Environments & evaluation:** multi-turn agent environments with tool calls and simulated users (incl. a vision-language planner under safety guidelines), task dataset construction, LLM-judge scoring and pairwise evaluation
- **Training & infrastructure:** multi-node distributed training with verl (Megatron-LM, vLLM), TRL, PyTorch, Transformers, Accelerate, Ray
- **Other:** Gymnasium, Stable Baselines, NumPy, Pandas; Python, C